

Demonstrating 3D Reconstruction to Modify 2D Laser Cutting Plans

Thijs Roumen
Hasso Plattner institute, University of
Potsdam, Germany
Thijs.Roumen@hpi.de

Conrad Lempert
Hasso Plattner institute, University of
Potsdam, Germany
Conrad.Lempert@hpi.de

Lukas Rambold
Hasso Plattner institute, University of
Potsdam, Germany
Lukas.Rambold@hpi.de

Muhammad Abdullah
Hasso Plattner institute, University of
Potsdam, Germany
Muhammad.Abdullah@hpi.de

Patrick Baudisch
Hasso Plattner institute, University of
Potsdam, Germany
Patrick.Baudisch@hpi.de

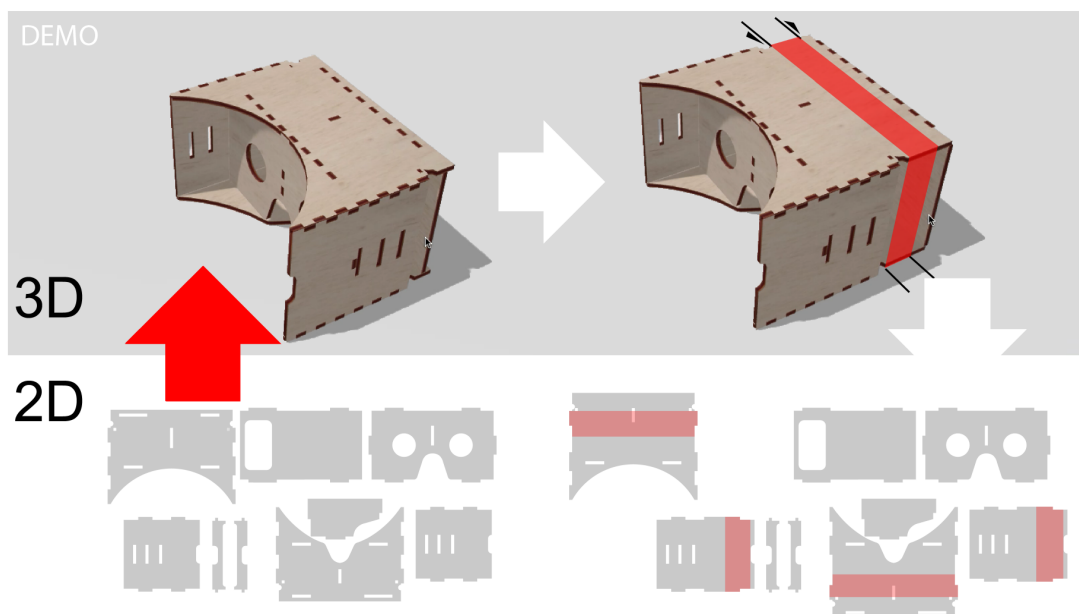


Figure 1: We demonstrate software tools to convert 2D cutting plans to 3D models either through user interaction (Assembler³[2]) or automatic using a heuristics-based beam search algorithm (autoAssembler [3]). This allows users to load the model into a 3D editor (kyub [1]), where they can apply parametric changes, and build on the work of others.

ABSTRACT

We demonstrate software tools to convert 2D cutting plans to 3D models either through user interaction (Assembler³[2]) or automatically using a heuristics-based beam search algorithm (autoAssembler [3]). This conversion allows users to make 3D manipulations (here using kyub [1]) to models for laser cutting, commonly shared in the form of 2D cutting plans. By sharing the models again in a 3D format, others can build on work and increase the complexity

Permission to make digital or hard copies of part or all of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).

CHI '22 Extended Abstracts, April 29–May 05, 2022, New Orleans, LA, USA

© 2022 Copyright held by the owner/author(s).

ACM ISBN 978-1-4503-9156-6/22/04.

<https://doi.org/10.1145/3491101.3519912>

of what is shared in the community. The resulting more advanced models and the ability to customize them to individual users' needs, increase the value of models for laser cutting shared online. In this demonstration, participants use the different software tools, and customize models to their liking.

CCS CONCEPTS

• **Human-centered computing** → Human computer interaction (HCI);

KEYWORDS

laser cutting, personal fabrication, reuse, remixing

ACM Reference Format:

Thijs Roumen, Conrad Lempert, Lukas Rambold, Muhammad Abdullah, and Patrick Baudisch. 2022. Demonstrating 3D Reconstruction to Modify

2D Laser Cutting Plans. In *CHI Conference on Human Factors in Computing Systems Extended Abstracts (CHI '22 Extended Abstracts)*, April 29–May 05, 2022, New Orleans, LA, USA. ACM, New York, NY, USA, 3 pages. <https://doi.org/10.1145/3491101.3519912>

1 INTRODUCTION

Models for laser cutting are hard to produce, they typically go through iterations of refinement and testing. Unsurprisingly, there is a strong community of enthusiasts who share such models. Remixing and customizing models has been a great driver for the field. In the context of laser cutting, typical modifications would be changing the material thickness and stretching the dimensions of the models aka “parametric changes”.

The most interesting laser cut models, consist of multiple plates that connect to one another using cutouts on the edges of the plates. These patterns form a joint together with the same (yet inverted) cutouts on another plate. Such joints allow users to make 3D models using 2D laser cutting plans.

Most models shared online, however, are in the form of 2D cutting plans (95% of over 200,000 models [2]), which makes parametric changes demanding, slow, and error prone.

2 3D RECONSTRUCTION

We demonstrate Assembler³ [2], a software tool that lets users convert 2D cutting plans into a 3D format, thus allowing users to make parametric changes more efficiently on that 3D format. Assembler³ is a five-step algorithm that automatically detects plates, joints, material thickness and pairs up joints using a hash table. It then presents the plates to users to puzzle them together into a 3D model.

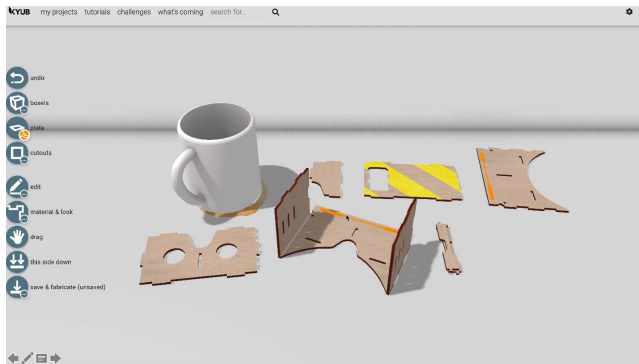


Figure 2: In assembler3, users manually reconstruct the models after the algorithm automatically (1) detected plates vs scrap, (2) found joint patterns on these plates, (3) used these joints to vote for a material thickness, and (4) paired up joints in a table.

We furthermore demonstrate an extension to this work called autoAssembler [3], a software tool that converts 2D cutting plans to a 3D format *automatically*, as illustrated by Figure 3. Since the search space is exponential in the number of joints, exhaustive search is impractical for any non-trivial model. AutoAssembler thus pursues only the most “promising” subset of candidates (beam

search). It considers candidates as promising if they (1) contain no intersecting plates, (2) fit into a small bounding box, (3) use plates whose joints fit together well, (4) the plates do not add many unpaired joints, (5) make use of constraints posed by other plates, and (6) conform to symmetry axes of the plates.

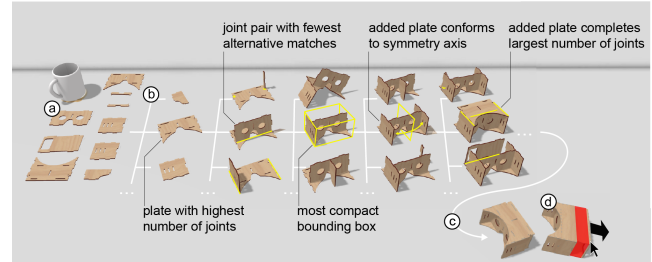


Figure 3: AutoAssembler uses a heuristics-based beam search algorithm to automatically convert 2D laser cutting plans to 3D models.

3 INTERACTIVITY

At the CHI demo we will demonstrate both Assembler³ and autoAssembler, this will give the audience a chance to see how Assembler³ can handle almost all laser-cut models given that their plates are joined with laser-cut features (100/105 in our technical evaluation). And at the same time, they can see the performance gain provided by autoAssembler, albeit on a subset of models. In both cases, the audience can then use kyub to make parametric changes to the reconstructed models.

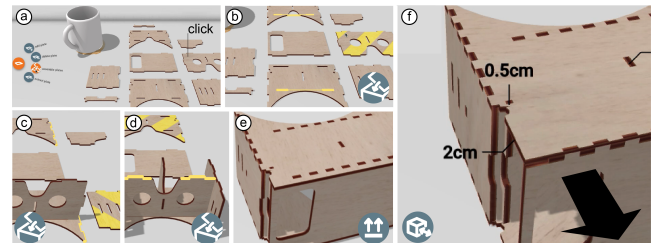


Figure 4: The user experience of Assembler³: using the assemble tool, users click together the 2D plates into a 3D model and then modify it.

We have two displays to show the interaction and ideas to the waiting audience, as well as two where they can make experience either autoAssembler or Assembler³ themselves. We provide the audience with a series of models so they have some choice but don’t end up spending time finding models or loading models that would take a long time to reconstruct. We will make sure to have a local install of kyub as a back-up in case there should be internet issues at the venue. We will furthermore bring two posters (one for each system) to explain additional details about the underlying algorithms for the passive audience to enjoy.

We provide the virtual audience with a comparable experience; they have the advantage that they do not need to wait to try the

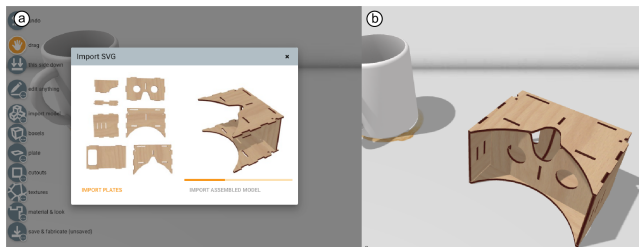


Figure 5: The user experience of autoAssembler is much shorter, giving the audience more time to modify the resulting model or learn about the underlying heuristics of autoAssembler.

demo as we will deploy a kyub install that has both autoAssembler and Assembler³ configured in it. We welcome them and explain in a short intro and demonstration what the audience is about to experience before letting them go ahead and try it out. We will provide them with a set of models like the in-person experience, but to their liking they can try out their own models as well.

4 CONCLUSION

We demonstrate Assembler³ [2] and autoAssembler [3], two software tools that bring the field closer to transitioning from 2D cutting

plans to 3D models, thereby providing an improved format for storing and sharing laser cut models. The increased ability for users to make parametric changes to the resulting models, will make it easier to customize models to their needs. It will furthermore allow more users to contribute and advance existing models, which in the long run may progress the laser cutting community as they create and share better and more complex models.

REFERENCES

- [1] Patrick Baudisch, Arthur Silber, Yannis Kommana, Milan Gruner, Ludwig Wall, Kevin Reuss, Lukas Heilman, Robert Kovacs, Daniel Rechltz, and Thijs Roumen. 2019. Kyub: a 3D Editor for Modeling Sturdy Laser-Cut Objects. In *2019 CHI Conference on Human Factors in Computing Systems Proceedings (CHI 2019)*, May 4–9, 2019, Glasgow, Scotland, UK. ACM, New York, NY, USA. DOI: <https://doi.org/10.1145/3290605.3300796>
- [2] Thijs Roumen, Yannis Kommana, Ingo Apel, Conrad Lempert, Markus Brand, Erik Brendel, Laurenz Seidel, Lukas Rambold, Carl Goedecken, Pascal Crenzin, Ben Hurdlehey, Muhammad Abdullah and Patrick Baudisch. 2021. Assembler³: 3D reconstruction of laser-cut models. In *CHI Conference on Human Factors in Computing Systems (CHI '21)*, May 8–13, 2021, Yokohama, Japan. ACM, New York, NY, USA, 18 pages. <https://doi.org/10.1145/3411764.3445453>
- [3] Thijs Roumen, Conrad Lempert, Ingo Apel, Erik Brendel, Markus Brand, Laurenz Seidel, Lukas Rambold, and Patrick Baudisch. 2021. autoAssembler: Automatic Reconstruction of Laser-Cut 3D Models. In *The 34th Annual ACM Symposium on User Interface Software and Technology (UIST '21)*, October 10–14, 2021, Virtual Event, USA. ACM, New York, NY, USA, 11 pages. <https://doi.org/10.1145/3472749.3474776>